sesa lab
SOFTWARE ENGINEERING
SALERNO

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Sustainable Software Engineering - Course Introduction



Gemma Catolino | 2026-2027

Sustainable Software Engineering: Teacher



Gemma Catolino
gcatolino@unisa.it

Office Hours

Monday: 11:00 - 13:00

Tuesday: 09:00 - 11:00

Please send an email at least one business day before.
The meeting can also be held using MS Teams.

Sustainable Software Engineering: General Information

Teacher	Gemma Catolino
Lectures Schedule	Monday: 13.30-15.30 Room F5 Tuesday: 11.00-13.00 Lab P13
E-learning:	https://elearning.informatica.unisa.it/el-platform/course/view.php?id=1422

Sustainable Software Engineering: Motivation behind this course



Environmental impact of software
The ICT sector accounts for an estimated 2–3% of global CO_2 emissions — roughly the same as the airline industry.

Rising demand for sustainable software
Tech giants such as Google, Microsoft, and Meta have pledged to achieve carbon neutrality by 2030.

Economic advantages
Embracing sustainable development practices can lower operating costs while improving energy efficiency.

Developers' social responsibility
Every line of code carries an environmental footprint.

Sustainable Software Engineering: Goal of this course

Provide a solid understanding of sustainability in software engineering, focusing on energy efficiency, environmental, social, and economic impact.



Energy optimization
Teach how to develop software that minimizes energy consumption and reduces environmental impact.

Long-term costs
Design solutions that are both cost-effective and sustainable over time, lowering maintenance expenses.

Social impact
Train developers to create technologies that are inclusive and socially responsible.

Regulatory compliance
Prepare to address the growing regulations on sustainability and accountability in software.

Sustainable Software Engineering: Learning Outcomes

Knowledge & Understanding

Apply principles and tools to optimize software for sustainability.

Applying Knowledge

Design sustainable applications across the software lifecycle wth an holistic approach.

Judgement

Make responsible design decisions considering technical, ethical, social, and environmental factors.

Communication

Clearly justify and present sustainable solutions to technical & non-technical audiences.

Learning Skills

Stay updated with emerging trends and apply sustainable practices in professional careers.

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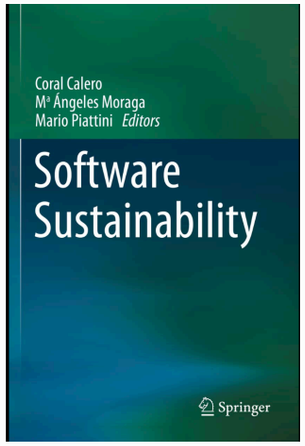
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M7. Case studies / Project	Review of real-world cases in green software and inclusivity.

Sustainable Software Engineering: Tentative Schedule

Modulo	Data	Orario	Aula	Topic
Introduzione alla sostenibilità nel software engineering	22/09/2026	11:00 - 13:00	P13	Course Introduction and Schedule + Sustainability 101
	28/09/2026	13:30 - 15:30	F5	Sustainability (Three Sides) 101
Efficienza energetica nel software	29/09/2026	11:00 - 13:00	P13	Energy Metrics and the Measurement Procedure
	05/10/2026	13:30 - 15:30	F5	Measuring Energy Consumption
	06/10/2026	11:00 - 13:00	P13	Energy Patterns Atelier
Efficienza energetica nel software/Pattern e strumenti per la sostenibilità	12/10/2026	13:30 - 15:30	F5	[LAB] A Tutorial about Green Software Tools
	13/10/2026	11:00 - 13:00	P13	Performance Testing
	15/10/2026	11:00 - 13:00	P6	[LAB] Unit-level performance testing: JMH, JU2JMH, AMBER
	16/10/2026	11:00 - 13:00	P6	[LAB] System-level performance testing: JMeter
Etica, Fairness e Giustizia sociale nel software	19/10/2026	13:30 - 15:30	F5	Social Sustainability
	20/10/2026	11:00 - 13:00	P13	Fairness in SSE
	26/10/2026	13:30 - 15:30	F5	Accountability, Transparency & Governance in SSE
	27/10/2026	11:00 - 13:00	P13	[LAB] Tool for measuring and mitigating Fairness
Ruolo degli sviluppatori e fattori umani nella sostenibilità	02/11/2026	13:30 - 15:30	F5	Human Factors and Socio-Technical Sustainability in Software Engineering
	03/11/2026	11:00 - 13:00	P13	Organizational Culture and Team Well-Being
	09/11/2026	13:30 - 15:30	F5	Social Debt and Community Smells
	10/11/2026	11:00 - 13:00	P13	[LAB] CADOCS, GUIDO, AND CODECITY LAB
Casi di Studio/Progetto	23/11/2026	13:30 - 15:30	F5	Role Play Community Smells
Valutazione economica, etica e sociale delle scelte di sviluppo	24/11/2026	11:00 - 13:00	P13	Total cost ownership and technical debt
	30/11/2026	13:30 - 15:30	F5	Software licensing
	01/12/2026	11:00 - 13:00	P13	[LAB] Fossa Licence
Casi di Studio/Progetto	07/12/2026	13:30 - 15:30	F5	GUEST LECTURE -TBD
	14/12/2026	13:30 - 15:30	F5	GUEST LECTURE -TBD
	15/12/2026	11:00 - 13:00	P13	Project Discussion

! **On October 15 and 16, we will take over Prof. Di Nucci's teaching hours for the Software Dependability course, as he will be away attending a conference.** !

Sustainable Software Engineering: Materials

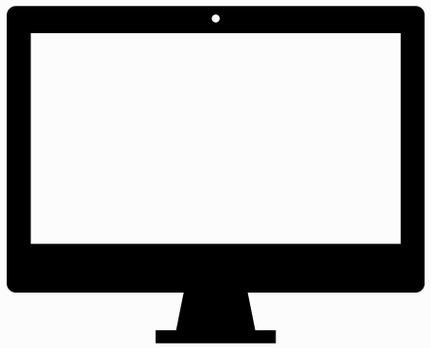


Calero, Coral, Mª Ángeles Moraga, and Mario Piattini. "Introduction to software sustainability." Software Sustainability. Cham: Springer International Publishing, 2021. 1-15.



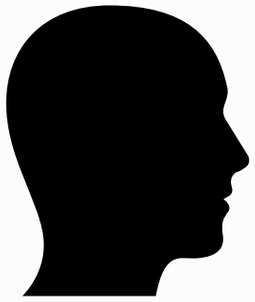
Slides provided by the teacher

Sustainable Software Engineering: Final Evaluation



Project

It will be the core of the exam. The grade will account for 50% of the final mark. The project must be presented and passed before the oral exam.



Oral Exam

It will cover all aspects of the course and will consist of 3–4 questions.



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